

ungroupGraphic [SimTalk]

Ungroups the designated group of graphics of the graphic designated by <Path>.

Remarks

Assigning the method deactivates graphic inheritance.

You cannot assign the method if the graphic is an automatically generated graphic group or if the graphic is located in an automatically generated graphic group.

Syntax

```
<Path>.ungroupGraphic
```

Type

Method

Examples

```
MyFrame._3D.getGraphic("default", [1,2,3]).ungroupGraphic
```

```
MyFrame._3D.getGraphic("deco", [1,2,3]).ungroupGraphic
```

See also

[Ungroup Group of Graphics](#)

[groupGraphics \[SimTalk\]](#)

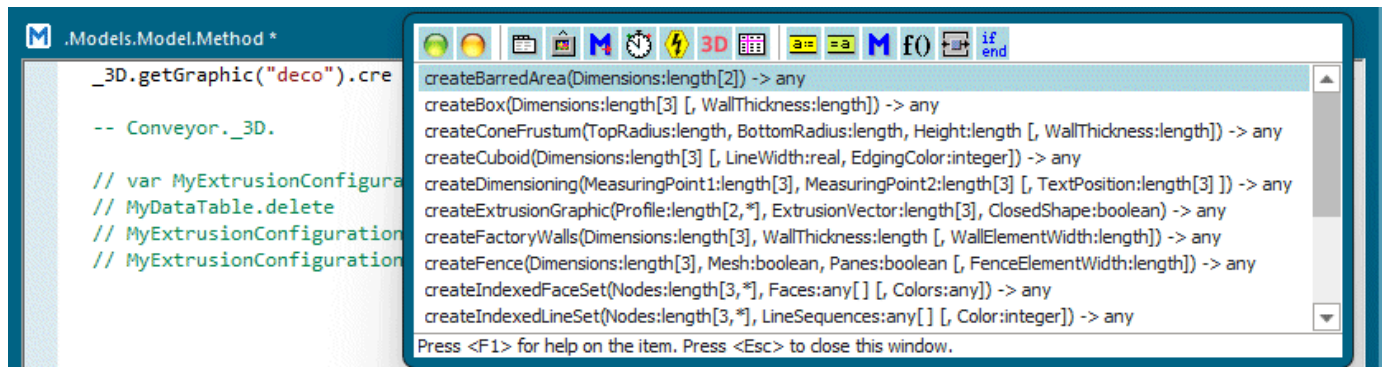
[_3D.getGraphic \[SimTalk\]](#)

Accessing Methods of Graphic Groups

__Accessing Methods of Graphic Groups

SimTalk provides the *methods* listed in the table of contents to the left for accessing graphic groups.

You can view and access the **methods** for accessing graphic groups by clicking **Auto Complete** on the **Edit ribbon tab** of the *Method Editor*.



See also

[Auto Complete](#)

[Edit 3D Properties \[dialog\]](#) > [Graphics \[tab\]](#) > [Graphic Groups \[described\]](#)

_3D.addGraphicGroup [SimTalk]

Adds the specified graphic group to the object designated by <Path>.

Remarks

Assigning the method deactivates graphic inheritance.

Type

Method

Syntax

```
<Path>._3D.addGraphicGroup(GraphicGroupName:string, Visible:boolean[,  
Internal:boolean:=false, Locked:boolean:=false])
```

Parameters

You can specify the following parameters:

- The parameter *GraphicGroupName* of data type *string* designates the name of the graphic group.
- The parameter *Visible* of data type *boolean* sets if the graphic group will be visible (*true*) or not visible (*false*).
- The optional parameter *Internal* of data type *boolean* sets if the graphic group will be internal (*true*) or external (*false*).

Default Value of the Parameter

The default value is *false*.

- The optional parameter *Locked* of data type *boolean* sets if the graphic group will be locked (*true*) or not (*false*).

Default Value of the Parameter

The default value is *false*.

Example

```
MyStation._3D.addGraphicGroup("MyGraphicGroup", true, true, false)  
  
// new source code, is the same as the source code below  
var graphicgroup := _3D.addGraphicGroup("MyGraphic", true, false)  
// previous source code  
_3D.addGraphicGroup("MyGraphic", true, false)  
var graphicgroup := _3D.getGraphic("MyGraphic")
```

[_3D.addGraphicGroup \[SimTalk\]](#)

See also

[_3D.deleteGraphicGroup \[SimTalk\]](#)

[Edit 3D Properties \[dialog\]](#) > [Graphics \[tab\]](#) > [Add \[graphic group\]](#)

[_3D.deleteGraphicGroup \[SimTalk\]](#)

Deletes the specified graphic group from the object designated by <Path>.

Remarks

Assigning the method deactivates graphic inheritance.

Type

Method

Syntax

```
<Path>._3D.deleteGraphicGroup(GraphicGroupName:string) → string
```

Parameter

The parameter *GraphicGroupName* of data type *string* designates the name of the graphic group.

Return Value

The return value has the data type *string*.

Example

```
MyStation._3D.deleteGraphicGroup("MyGraphicGroup")
```

See also

[_3D.addGraphicGroup \[SimTalk\]](#)

[Edit 3D Properties \[dialog\]](#) > [Graphics \[tab\]](#) > [Delete \[button\]](#) - graphic group

[_3D.deleteGraphicGroupContent \[SimTalk\]](#)

Deletes the graphics from the specified graphic group from the object designated by <Path>.

Remarks

Assigning the method deactivates graphic inheritance.

Type

Method

Syntax

```
<Path>._3D.deleteGraphicGroupContent([GraphicGroupName:string])
```

Parameter

The optional parameter *GraphicGroupName* of data type *string* designates the name of the graphic group.

Do not specify the name to delete the content of the **default** graphic group.

If you try to delete the content of an automatically generated default graphic group, this attempt will fail without an error message.

Specify a name to open the *Method Debugger*.

Example

```
MyFrame._3D.deleteGraphicGroupContent("deco")
```

Accessing Attributes and Read-Only Attributes of Graphic Groups

[__Accessing Attributes and Read-Only Attributes of Graphic Groups](#)

SimTalk provides the *attributes* and *read-only attributes* listed in the table of contents to the left for accessing **graphic groups**.

You can view and access the **attributes** for accessing graphic groups by clicking **Auto Complete** on the **Edit ribbon tab** of the *Method Editor*.

See also

[Auto Complete](#)

Edit 3D Properties [dialog] > Graphics [tab] > Graphic Groups [described]

_3D.ExternalGraphicGroups [SimTalk]

Sets the list of all external graphic groups of the *Frame* designated by <Path>.

Remarks

Visible external graphic groups are shown in 3D windows that opened a *Frame* which contains the designated *Frame* that owns these graphic groups.

Note

Assigning the attribute deactivates graphic inheritance of the *Frame* designated by <Path>.

Type

Attribute

Syntax

```
<Path>._3D.ExternalGraphicGroups:string[]
```

Assignment Value

You can assign an *array* of data type *string*.

Example

```
.Models.Model._3D.ExternalGraphicGroups := ["default", "alternative"]  
var a : string[] := .Models.Model._3D.ExternalGraphicGroups
```

See also

Edit 3D Properties [dialog] > Graphics [tab] > Internal [graphic group]	Internal [SimTalk] - graphic group
Edit 3D Properties [dialog] > Graphics [tab] > External [graphic group]	External [SimTalk] - graphic group
Edit 3D Properties [dialog] > Graphics [tab] > Visible [graphic group]	Visible [SimTalk] - graphic group
_3D.InternalGraphicGroups [SimTalk]	_3D.VisibleGraphicGroups [SimTalk]

[_3D.GraphicGroupNames \[SimTalk\]](#)

Returns a list of all graphic groups that exist for the object designated by <Path>.

Type

Read-only attribute

Syntax

```
<Path>._3D.GraphicGroupNames → string[]
```

Return Value

The return value is an *array* of data type *string*.

Example

```
print .Models.MyEnginePlant.Station._3D.GraphicGroupNames  
// might, for example, return [default, mygraphicgroup1, MyGraphicGroup2]
```

See also

[Edit 3D Properties \[dialog\]](#) > [Graphics \[tab\]](#) > [Graphic Groups \[described\]](#)

_3D.InternalGraphicGroups [SimTalk]

Sets the list of all internal graphic groups of the *Frame* designated by <Path>.

Remarks

Visible internal graphic groups are shown in 3D windows that opened the designated *Frame* that owns these graphic groups.

Note
Assigning the attribute deactivates graphic inheritance of the *Frame* designated by <Path>.

Type

Attribute

Syntax

```
<Path>._3D.InternalGraphicGroups:string[]
```

Assignment Value

You can assign an *array* of data type *string*.

Example

```
.Models.Model._3D.InternalGraphicGroups := ["deco1", deco2]  
var a : string[] := .Models.Model._3D.InternalGraphicGroups
```

See also

Edit 3D Properties [dialog] > Graphics [tab] > Internal [graphic group]	Internal [SimTalk] - graphic group
Edit 3D Properties [dialog] > Graphics [tab] > External [graphic group]	External [SimTalk] - graphic group
Edit 3D Properties [dialog] > Graphics [tab] > Visible [graphic group]	Visible [SimTalk] - graphic group
_3D.ExternalGraphicGroups [SimTalk]	_3D.VisibleGraphicGroups [SimTalk]

_3D.LockedGraphicGroups [SimTalk]

Sets the list of all locked graphic groups of the object designated by <Path>.

Remarks

Locking graphic groups prevents you from selecting this graphic group in the 3D windows which are opened for this object. This makes, for example, sense for the graphic of a factory building which might otherwise be selected inadvertently.

Note

Assigning the attribute deactivates graphic inheritance.

Type

Attribute

Syntax

```
<Path>._3D.LockedGraphicGroups:string[]
```

Assignment Value

You can assign an *array* of data type *string*.

Example

```
.Models.Model.Station1._3D.LockedGraphicGroups := ["Group1, Group2"]  
var a: string[] := .Models.Model.Station1._3D.LockedGraphicGroups
```

See also

Edit 3D Properties [dialog] > Graphics [tab] > Locked [graphic group]	_3D.InternalGraphicGroups [SimTalk]
Locked [SimTalk] - graphic group	_3D.VisibleGraphicGroups [SimTalk]

_3D.VisibleGraphicGroups [SimTalk]

Sets the graphic groups of the object designated by <Path> that will be visible.

Remarks

Not visible graphic groups are not shown in any 3D window.

This makes, for example, sense for objects that you want to visualize in different development stages.

The setting **Visible** allows you to select one or more of multiple graphic groups for the current representation of the object.

Note

Assigning the attribute deactivates inheritance of the setting **Visible** of all graphic groups of the designated *Frame*.

Type

Attribute

Syntax

```
<Path>._3D.VisibleGraphicGroups:string[]
```

Assignment Value

You can assign an *array* of data type *string*.

Example

```
MyStation._3D.VisibleGraphicGroups := ["default", "deco"]
```

See also

Edit 3D Properties [dialog] > Graphics [tab] > Internal [graphic group]	Internal [SimTalk] - graphic group
Edit 3D Properties [dialog] > Graphics [tab] > External [graphic group]	External [SimTalk] - graphic group

Edit 3D Properties [dialog] > Graphics [tab] > Visible [graphic group]	Visible [SimTalk] - graphic group
	_3D.VisibleGraphicGroups [SimTalk]

External [SimTalk] - graphic group

Sets if the graphic group designated by <Path> is to represent its owning *Frame* toward the outside (`true`) or not (`false`).

Remarks

Plant Simulation shows an external graphic group in 3D windows that opened an object which contains the *Frame* which owns the graphic group, but only if the graphic group is set to *Visible* at the same time.

Plant Simulation also shows a visible external graphic group, even it is not internal at the same time, in 3D windows that opened the *Frame* which owns the graphic group and shows external graphic groups marked as *Visible*.

Note

Assigning the attribute deactivates graphic inheritance of the owning *Frame*.

Type

Attribute

Syntax

```
<Path>.External: boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
print _3D.getGraphic("MyGraphicGroup").External  
_3D.getGraphic("MyGraphicGroup").External := true
```

See also

[Visible \[SimTalk\] - graphic group](#)

[Graphic Groups \[described\]](#)

[_3D.getGraphic \[SimTalk\]](#)

[Show External Graphic Groups](#)

Generated [SimTalk] - graphic group

Returns if the graphics of the graphic group designated by <Path> are generated automatically or not.

Remarks

If the graphic group is generated, you can neither add graphics to it nor delete graphics from it. You can edit its visibility settings though.

Type

Read-only attribute

Syntax

```
<Path>.Generated: boolean
```

Return Value

The return value has the data type *boolean*.

Example

```
print Station._3D.getGraphic("MyGraphicGroup").Generated
```

See also

[Graphic Groups \[described\]](#)

[_3D.getGraphic \[SimTalk\]](#)

Internal [SimTalk] - graphic group

Sets if the graphic group designated by <Path> is to decorate its owning *Frame* inward (*true*) or not (*false*).

Remarks

Plant Simulation shows an internal graphic group in 3D windows that opened the *Frame*, which owns the graphic group, but only if the graphic group is set to *Visible* at the same time.

Note

Assigning the attribute deactivates graphic inheritance of the owning *Frame*.

Type

Attribute

Syntax

```
<Path>.Internal: boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
print _3D.getGraphic("MyGraphicGroup").Internal  
_3D.getGraphic("MyGraphicGroup").Internal := true
```

See also

[Visible \[SimTalk\] - graphic group](#)

[Graphic Groups \[described\]](#)

[_3D.getGraphic \[SimTalk\]](#)

[Show External Graphic Groups](#)

Locked [SimTalk] - graphic group

Sets if the graphic group designated by <Path> is locked (`true`) or not (`false`).

Remarks

Locking a graphic groups prevents you from selecting this graphic group in the 3D windows which are opened for this object.

This makes, for example, sense for the graphic of a factory building which might otherwise be selected inadvertently.

Type

Attribute

Syntax

```
<Path>.Locked: boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
print Station._3D.getGraphic("MyGraphicGroup").Locked  
Station._3D.getGraphic("MyGraphicGroup").Locked := false
```

See also

[Graphic Groups \[described\]](#)

[_3D.getGraphic \[SimTalk\]](#)

Visible [SimTalk] - graphic group

Sets if the graphic group designated by <Path> is visible (`true`) or not (`false`).

Remarks

Plant Simulation does not show a not visible graphic group in any 3D window.

This makes, for example, sense for objects that you want to visualize in different development stages. The setting *Visible* allows to select one or more of multiple graphic groups for the current representation of the object.

Note

Assigning the attribute *Visible* deactivates inheritance of the visible setting of all graphic groups of the owning object.

Type

Attribute

Syntax

```
<Path>.Visible:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
print Station._3D.getGraphic("MyGraphicGroup").Visible  
Station._3D.getGraphic("MyGraphicGroup").Visible := false
```

See also

[Graphic Groups \[described\]](#)

[Visible \[graphic group\]](#)

[Visible \[context menu\]](#)

[_3D.getGraphic \[SimTalk\]](#)